FINANCIAL MANAGEMENT

Suggested Answers
March-April 2024

Answer to the Question# 1(a):

Capital allowances

	Tk.	Tax saved @ 27% Tk.	Timing
Cost	5,000		
Year 0 WDA (18%)	(900)	243	t0
	4,100		
Year 1 WDA	(738)	199	t1
	3,362		
Year 2 WDA	(605)	163	t2
	2,757		
Year 3 WDA	(496)	134	t3
	2,261		
Year 4 sales proceeds	-		
Balance allowances	2,261	610	t4

Answer to the Question# 1(b):

Investment decision

Cash flows	t0 Tk.	t1 Tk.	t2 Tk.	t3 Tk.	t4 Tk.
Purchase of machine	(5,000)				
Tax saved through WDAs	243	199	163	134	610
Net revenue		3,000	3,000	1,000	1,000
Tax on net revenue		(810)	(810)	(270)	(270)
	(4,757)	2,389	2,353	864	1,340
Discount factor	1.000	0.909	0.826	0.751	0.683
Present value	(4,757)	2,172	1,944	649	915

NPV = + Tk. 923
Therefore, the project should be accepted.

Answer to the Question# 1(c):

Different timing of initial purchase

PV of tax savings	Tk.
Tk. 243 x 1.000 + Tk. 199 x 0.909 + Tk. 163 x 0.826 +Tk. 134 x 0.751 + Tk. 610 x 0.683	1,076
Hence, PV if delayed by one year Tk. 1,076 x 0.909	978
Difference	98

Hence, new NPV = Tk. (923 - 98)
Tk. 825
This means the project is still worthwhile.

Wisdom - fine if there is a balancing charge (delay it!)
- not so if there is a balancing allowance

Answer to the Question# 2(a) (i):

Part A: Credit Policy and Agency Subscription

i) Factors Affecting a Company's Credit Policy

1. **Customer Creditworthiness:**
   - Assessing the ability and reliability of customers to pay back the credit extended. Companies often use credit scores, credit history, and financial statements for this purpose.
2. **Market Conditions:**
   - Economic conditions, industry trends, and market competition can influence how liberal or conservative a company's credit policy should be.
3. **Competition:**
   - Consideration should be given to the terms of credit your competitors are offering
4. **Risk Tolerance:**
   - The company's willingness to take on the risk of non-payment. This is influenced by its financial strength and strategic objectives.
5. **Credit Terms:**
   - The specific terms of credit such as the length of the credit period, interest rates on overdue payments, and discounts for early payments can affect the credit policy.

Answer to the Question# 2(a) (ii):

Description	Rate			BDT
Initial charge				200,000
Additional charge	5,000	X	12,000 sales	60,000,000
Cost of the subscription				**60,200,000**
Costs of Bad Debts without any subscription				
Bad debts	8%			
Cost of sales	12,000	X	65,000	780,000,000
Bad debts				**62,400,000**
Savings under the Subscription				**2,200,000**

The company should subscribe to the credit agency.

Answer to the Question# 2(b) (i):

List TWO activities that increase cash and TWO activities that decrease cash.

Activities that Increase Cash:

1. **Collecting Receivables:**
   - Faster collection of accounts receivable increases cash inflows and improves liquidity.
2. **Selling Inventory:**
   - Selling finished goods increases cash inflows from operations.

Activities that Decrease Cash:

1. **Purchasing Inventory:**
   - Buying raw materials or work-in-progress inventory requires cash outflows.
2. **Paying Creditors:**
   - Paying off accounts payable results in cash outflows.

Answer to the Question# 2(b) (ii):

Description	Computations			Days
Raw materials stock period	54,000,000	/	1,068,494	51
(-) Creditor period	46,000,000	/	1,068,494	(43)
Work in progress period	39,500,000	/	1,178,083	34
Finished inventory turnover	35,500,000	/	1,178,083	30
Debtor-conversion period	35,000,000	/	1,287,672	27
Cash conversion cycle				**98**

Answer to the Question# 3(a):

Cost of capital calculation

Cost of equity (Ke)

Using the dividend valuation model, Ke = [Do(1+g) / Po]+g
Assuming the dividend growth over the last five years is a good indicator of shareholders' expectations regarding the future:
g = (25.5/21.5)^1/5 - 1
= 3.71%
Thus Ke= 25.5 (1.0371/335-25.5)+0.0371
= 12.25%

MV of share = 575,000 x Tk. 309.5 = Tk. 177,962,500

Cost of Debt (Kd)
The Kd will equal the IRR of the cash flows from the company's perspective
The debt will be redeemed in one year, so it is possible to calculate the cost exactly.

Consider a bond with nominal value Tk. 100.

Time	Narrative	Cash Flow	Discount factor	Present value
0	MV (ex int)	93	1.000	92.72
1	R = Interest	(109)	0.851	(92.72)
	NPV			0.00

The t = 1 discount figure is 1/(1+Kd) which must equal 0.851 to give a NPV of zero.
Thus 1+Kd = 1/0.851 = 1.1751 giving pre-tax Kd = 17.51%
Post tax cost of debt = 17.51 (1-0.27) =12.78%
The MV of the debt = Tk. 100 million x 92.72% = Tk. 92,720,000 ex interest

WACC

The WACC normally includes long-term finance only. In the case of Fiz Ltd the debt is to be redeemed in one year's time. Thus the debt should be included in the WACC if it is assumed that more debt will be raised in the near future to replace it. (if not, then the WACC is the cost of equity of 12.25%

WACC = (12.25 x 177.96 + 12.78 x 92.72) / 270.68
= 12.43%

Answer to the Question# 3(b):

Reliability of WACC

The WACC determined may not be totally reliable for the following reasons:

1. **Market Conditions**: The WACC is based on current market conditions which may change in the future, especially with the investment in a new region like the Middle East.
2. **Risk Differences**: The risk profile of the new investment in the Middle East may differ from the existing operations in Bangladesh. The WACC calculated is based on the company's current risk, which might not fully capture the risks associated with the new venture.
3. **Assumptions**: Several assumptions were made, such as the constant growth rate of dividends and the use of market values which may fluctuate.
4. **Tax Rate Stability**: The assumption that the corporate tax rate will remain constant at 27% may not hold true in the future, affecting the after-tax cost of debt.
5. **Currency Risk**: The new investment will be subject to currency risk if revenues and costs are in different currencies. This is not accounted for in the WACC calculation.
6. **Cost of Equity Calculation**: The dividend growth model used for calculating the cost of equity assumes a stable and perpetual growth rate which may not be realistic.

These factors suggest that while WACC provides a useful benchmark, it should be adjusted for the specific risks and conditions of the new investment.

Suitability
The WACC has been calculated to use as a discount rate for appraising the new overseas venture. To use the existing WACC relies on the following assumptions:
- Gearing is constant
- The project has the same business risk as the company's existing activities
- The project is small

The figure calculated is not suitable for the following reasons:
- It is implied that the project will be financed out of retained cash reserves. This will not preserve the current gearing of the company.
- The overseas venture involves a different market with different systematic risk to the bulk of existing activities.
- The project is a major undertaking.

Answer to the Question# 4(a):

No, it is not advisable to use short term financing to finance its entire current assets. The magnitude of currents assets increases or decreases over time and also with respect to level of activity. However, there is always a minimum level of current assets which is continuously required by the firm to carry on its business operations. This is referred as permanent working capital. The permanent working capital to be treated as

fixed assets for financing decision, and so it is advisable to use long term funds for financing of permanent level of current assets.

If the firm uses short term financing to finance its current assets entirely, then it runs the risk of renewing borrowings again and again. This continued financing exposes the firm to certain risks like to borrow during stringent credit, and/or to borrow at most inconvenient terms, etc.

Answer to the Question# 4(b)(i):

Cost of debt:
Net proceeds = Par value (Tk. 1,000) - Discount (Tk. 30) - Flotation costs (Tk. 30) = Tk. 940.
The before-tax cost of debt is the interest rate that makes the following equation hold:

$$B_n = \frac{C}{(1+r_d)^2} + \frac{C}{(1+r_d)^2} + \ldots + \frac{C}{(1+rd)^n} + \frac{M}{(1+r^d)^n}$$

where:

B_n = Net proceeds from bond sale (Tk. 940), M = Par value in taka (Tk. 1,000)
C = Coupon payment each period (8%), rd = yield to maturity (before-tax cost of debt)
N- periods to maturity (20)

$$Tk.940 = \frac{Tk.80}{(1+r_d)^2} + \frac{Tk.80}{(1+r_d)^2} + \ldots + \frac{Tk.80}{(1+r_d)^{20}} + \frac{Tk.1{,}000}{(1+r_d)^{20}} = 8.64\%$$

The after-tax cost of debt is r_d x (1 – T), where T IS the tax rate. With a 21% tax rate, the after-tax cost of debt is 0.0864 × (1- 0.21) = 0.0683 = 6.83%.

Answer to the Question# 4(b)(ii):

Cost of preferred stock:

The cost of new preferred stock (r_p) is given by ($D_p \div N_p$), where D_p is the expected perpetual annual dividend on preferred stock and N_p is net proceeds from selling new preferred stock. N_p = Sales price of new preferred stock - Flotation costs = Tk. 95 – Tk. 5 = Tk. 90. Preferred dividends are 8% of par value (Tk 95) or Tk. 7.60, so r_p = Tk. 7.60 ÷ Tk. 90 = 8.44%.

Answer to the Question# 4(b)(iii):

Cost of retained earnings:

The price of Lany's common stock (P_0) is Tk. 90, dividend growth has been 6% per year and is expected to continue, and the next expected dividend is Tk. 7. When a firm uses retained earnings, there are no flotation or underpricing costs, so the cost of common equity is simply the required return on common stock, rs = ($D_1 \div P_0$) + g = (Tk. 7 ÷ Tk. 90) + 0.06 = 0.0778 + 0.0600 = 0.1378= 13.78%.

Cost of new common stock: The cost of new common stock (r_n) is given by ($D_1 \div N_n$) - g, where N_n is net sales proceeds. Flotation costs are Tk. 5 per share, and new shares must be underpriced by Tk. 7 per share. Net proceeds = Price of common stock (net of underpricing) – Flotation costs = [(Tk.90 – Tk. 7) – Tk. 5] = Tk. 78, so r_n = (Tk. 7 ÷ Tk. 78) + 0.06 = 0.0897 + 0.0600 = 14.97%.

Answer to the Question# 4(b)(iv):

Type of Capital	Weights (%)	Cost	Weighted Cost
With retained earnings			
Long-term debt	0.30	6.83%	2.048%
Preferred stock	0.20	8.44%	1.689%
Common stock equity	0.50	13.78%	6.889%

	Sum (r_{WACC}) =10.6%*

Type of Capital	Weights (%)	Cost	Weighted Cost
With new common stock			
Long-term debt	0.30	6.83%	2.048%
Preferred stock	0.20	8.44%	1.689%
Common stock equity	0.50	14.97%	7.487%
			Sum (r_{WACC}) = 11.2%

Answer to the Question# 5(a):

Year	Items	Amount	PV Factor @12%	PV
0	Purchase of Machinery	(4,700,000)	1.00	(4,700,000)
01-04	Annual contribution margin (10,000-7,000)*800	2,400,000	3.037	7,288,800
01-04	Incremental Fixed cost (975,000-75,000-250,000)	(650,000)	3.037	(1,974,050)
4	Sales of Machinery	800,000	0.636	508,800

NPV of producing camera	1,123,550
NPV of selling the plant	1,300,000
Excess NPV of selling the plant	**176,450**

Answer to the Question# 5(b):

Sensitivity analysis

Initial outlay:

= $\frac{1,123,550}{4,700,000}$ X 100 = 23.91%

Discount rate:
Have to find IRR of the project at first.

Year	Items	Amount	PV Factor @12%	PV	PV Factor @23%	PV
0	Purchase of Machinery	(4,700,000)	1.00	(4,700,000)	1.00	(4,700,000)
01-04	Annual contribution margin (10,000-7,000)*800	2,400,000	3.037	7,288,800	2.448	5,875,200
01-04	Incremental Fixed cost (975,000-75,000-250,000)	(650,000)	3.037	(1,974,050)	2.448	(1,591,200)
4	Sales of Machinery	800,000	0.636	508,800	0.437	349,600
	NPV of producing camera			1,123,550		(66,400)

By using extrapolation, IRR can be found as:

12% + $\frac{1,123,550}{1,123,550+66,400}$ (23% - 12%)

= 12% + 10.38%

= 22.38% ≈ 22.3% (appx.)

Therefore sensitivity is ((22.39/12)-1) = 86.5%

Residual value:

$= \frac{1,123,550}{508,800}$ X 100 = 220.82%

Annual net operating cash flow:

(4,700,000) + 3.037 x + 508,800 = 0
X = Tk. 1,380,046

Sensitivity = $\frac{1,123,550}{(7,288,800-1,974,050)} \ X \ 100$ = 21.14%

Answer to the Question# 5(c):

Evaluation

Finding in (a)

After comparing the NPV of selling the product and selling the product, it is found that selling the patent immediately has Tk. 176,450 more NPV than the option of producing. Thus, based on this financial information only, the patent should be sold.

Finding in (b)

Initial outlay must increase by 24% to make the NPV zero (moderate risky).

Discount rate need to be increased by 86% to make the NPV zero (low risk).

Residual value need to be decreased by 220.82% to make NPV zero (low risk).

Annual net operating cash flow need to decrease by 21% to make NPV zero (moderate risk).

Therefore, based on the sensitivity overall risk of the estimated NPV is said to be low

Answer to the Question# 6(a):

Impact of the new project and related finance.

	Tk. In million
Current ex-div value of shares = 5 million x Tk. 5.00	25.00
NPV of the project	0.50
Increase in equity capital	0.50
Ex-div value of shares afterwards	26.00

This will be the ex-div value of the equity irrespective of how the money is raised. The cum-div value will obviously be affected if some of the dividend is retained, but the wealth of the

Requirement(i):

Retained earnings

Tk. In million		Ex - div value	+ Dividend	= Cum - div	
Before		25.00	2.50	27.50	Increase in shareholders
Capital: using dividend		0.50	(0.50)	-	weath is Tk. 0.5 million
Project outlay	(0.50)				Being NPV of the project
PV (inflow)	1.00				
NPV Project		0.50		0.50	
		26.00	2.00	28.00	

Requirement(ii):

Right Issue

		Ex - div value	+ Dividend	= Cum - div	
Before		25.00	2.50	27.50	Increase in shareholders wealth:
Capital:		0.50		0.50	Increase in value of equity: Tk. 1.0 million
Project outlay	(0.50)				Less: cost of new shares Tk. 0.5 million
PV (inflow)	1.00				
NPV Project		0.50		0.50	
		26.00	2.50	28.50	

In conclusion, the change in shareholders' wealth is the same, irrespective of how the new capital is injected.

Requirement(iii):

While the wealth of the shareholders in total is unaffected by the source of the new equity, the way that wealth is divided between the shareholders is influenced by the price chosen. The company therefore needs to consider carefully the asking price per share.

If these new shareholders are allowed to enter the company too cheaply, the increased number of shares will detract from the wealth of existing shareholders; if priced too highly they will refuse to subscribe.

If all of the gain is to go to the existing shareholders the ex-div value of their 5m shares must be Tk. 25.5m after accepting the project i.e Tk. 5.10 each. If the company wants to set a price of Tk. 5.10, they will need to issue Tk. 500,000 / Tk. 5.10 = 98,039 new shares.
Taking into account the NPV of the subsequent project, the value of the shares would change as follows.

Issuing new shares at a price of Tk. 5.10

	Shares	Price (Tk.)	Value of share (Tk.)
Amount issued	98,039	5.10	500,000
Currently in existence	5,000,000	5.00	25,000,000
			25,500,000
NPV of the project			500,000

5,098,039	5.10	26,000,000

The combined value of the shares would be Tk. 26 million. This would be shared over the 5,089,039 shares to give a share price of Tk. 5.10

	Shares	Price (Tk.)	Value of share (Tk.)	Gain (Tk.)
Amount issued	98,039	5.10	500,000	-
Currently in existence	5,000,000	5.10	25,500,000	500,000

Once all shares have moved to the equilibrium price of Tk. 5.10, it can be seen that all of the gain has in fact gone to the existing shareholders.

Answer to the Question# 6(b) (i):

To find NPV for the standard and custom plants, first recognize NPV is given by:

Present value of cash inflows from project —Upfront project cost.

The present value of cash inflows for the standard plant would be:

= PV (discount rate, number of periods, -cash inflows) = PV (0.169,10, -12.5) = Tk. 58.44 million

Project cost is Tk. 38.5 million, so NPVstandard Tk. 58.44 – Tk. 38.5 = Tk. 19.94 million.

The present value of cash inflows for the custom plant would be:

= PV (discount rate, number of periods, -cash inflows) = PV (0.169,10, -17.5) = Tk. 81.82 million
Project cost is Tk. 53.5 million, so NPVcustom= Tk. 81.82 – Tk. 53.5 = Tk. 28.32 million.

Both the standard and custom plants have positive NPVs and, therefore, are acceptable.

Answer to the Question# 6(b) (ii):

The breakeven annual cash inflow is the cash inflow that, for the given life of the project, upfront cost of the project, and cost of capital, makes project NPV equal zero. The cash inflow can be find out in the following manner:

= PMT (discount rate, number of periods, -present value),

where the upfront cost of the project is inserted for present value

For the standard plant, breakeven annual cash flow is: =PMT (0.169,10, -38.5) = Tk. 8.23 million

For the custom plant, breakeven annual cash flow is: =PMT (0.169,10,-53.5) = Tk 11.44 million.

Answer to the Question# 6(b) (iii):

To find the probability cash flows will meet/exceed the breakeven level, sum the probabilities for ranges greater than or equal to the specific range where breakeven cash flows lie for each project:

Cash flow range	STANDARD PLANT		CUSTOM PLANT	
	Probability of Cash Flow in given range	Cumulative probability	Probability of Cash Flow in given range	Cumulative probability
Above Tk. 20	0%	0%	10%	10%
Tk.17-Tk.20	0%	0%	15%	25%
Tk.14-Tk.17	5%	5%	20%	45%
Tk.11-Tk.14	25%	30%	25%	70%
Tk. 8-Tk. 11	60%	90%	15%	85%
Tk.5-Tk.8	10%	100%	10%	95%
Tk.0-Tk.5	0%	100%	5%	100%
Breakeven Cash Flow	Tk 8.23mn		Tk 11.44mn	

Answer to the Question# 6(b) (iv):

Relative project risk can be evaluated-in several ways. For example, cash flows for the standard plant are distributed relatively tightly around the breakeven level; there is a 60% chance annual cash flows will be Tk. 8 to Tk. 11 million and a 95% chance they will be between Tk. 5 and S14 million. For the custom plant, cash

flows could range from Tk. 0 to over Tk. 20 million, and the probability cash flows will be in the bucket containing the breakeven cash flows is only 25%. From another perspective, the probability that cash flows will lie below the breakeven level (obtained by summing probabilities for ranges below breakeven range) is 10% for the standard plant but 30% for the custom plant. From still another perspective, the probability that cash flows will clearly generate a positive NPV (obtained by summing probabilities for ranges above the breakeven range) is 30% for the standard plant but 45% for the custom plant.

---The End---